OFFICIAL PRODUCT DOSSIER

HireSense AI Platform Dossier

AI-Powered Resume Intelligence, Explainable Matching & Talent Acquisition SaaS

VERSION: v1.0.0 Production	**REPO: github.com/VaibhavS18S/hiredsense**	**TEST SUITE: 5/5 PASSED (100%)**
TECH STACK: MERN + Pluggable XAI	BRANCH: main (Synced)	**STATUS: Production Ready**

1. Executive Overview & Problem Statement

HireSense AI is a full-stack, enterprise-style talent acquisition platform built using the MERN stack (MongoDB, Express, React, Node.js) combined with Explainable AI (XAI) and natural language processing. The product addresses fundamental failures in modern recruitment software: black-box algorithms, opaque rejection criteria, and keyword-stuffing exploits.

Unlike generic ATS tools that act as silent gatekeepers, HireSense AI maintains a strict Human-in-the-Loop policy. The platform calculates match scores from verifiable evidence, extracts direct textual citations from resumes, highlights candidate skill gaps, and equips hiring teams with side-by-side comparisons, Kanban pipelines, and interview rubrics.

> **Design Philosophy: Distinctive Editorial Aesthetics**
>
> Designed with a warm off-white canvas (#F7F7F4), deep obsidian ink typography (#171A21), dark forest-green accents (#176B52), and sage surfaces (#

2. End-to-End Recruitment Workflow (6-Stage Pipeline)

The platform organizes candidate screening into a structured, automated six-stage lifecycle:

- **1. Upload & Ingestion: Batch upload supporting PDF, DOCX, and TXT files with SHA-256** duplicate detection.
- **2. Text Extraction: Robust streaming extraction via pdf-parse and mammoth, isolating** header, body, and contact.
- **3. Information Parsing: Determines tenure dates, educational credentials, phone** numbers, emails, and links.
- **4. Taxonomy Mapping: Extracts skills across 8 technical domains using a 500+ term** regex boundary engine.
- **5. XAI Scoring Engine: Calculates 4-part weighted score with verbatim textual** citations for every matched requirement.
- **6. Recruiter Review: Enables human verification, manual score adjustments, notes, and** pipeline stage movement.

3. Core Application Modules (Detailed Feature Breakdown)

Module A: Executive Recruitment Dashboard

The central command center provides real-time recruitment KPIs calculated directly from persistent database records: Active Job Openings, Total Candidates Processed, Review Backlog, Shortlist Conversion Rate, Average Time-to-Shortlist, and historical screening volume trends visualized with interactive Recharts area graphs.

Module B: Job Intelligence & Readiness Analyzer

Allows hiring teams to create, edit, duplicate, and analyze job descriptions. Includes an automated 0-100% Job Readiness Score that evaluates criteria clarity, role responsibilities, and mandatory competencies before publishing. A built-in Buzzword Detector flags non-informative clichÃ©s (e.g., "rockstar", "ninja", "guru") and suggests concrete competency definitions. Recruiters can also click "AI Enhance" to generate structured job criteria.

Module C: Smart Resume Processing Center

Equipped with a bulk drag-and-drop interface supporting PDF and DOCX documents up to 15MB. Calculates SHA-256 fingerprints to prevent duplicate processing, shows real-time multi-stage progress bars, and alerts recruiters if a scanned document or corrupt file requires optical character recognition.

Module D: Explainable AI Matching Engine & Evidence Extraction

Eliminates guesswork by decomposing candidate-job alignment into four normalized, auditable components with configurable administrative weights:

Scoring Dimension	Default Weight	Measurement Method	Evidence Provided
Skill Alignment	40%	Mandatory (85% base) + Preferre...	Verbatim resume quotes & citations
Semantic Fit	30%	Vector cosine similarity (TF-ID...	Distance score & semantic corre...
Relevant Experience	20%	Parsed employment years vs thre...	Timeline verification & job tenure
Qualifications	10%	Degree hierarchy (BS, MS, PhD) ...	Degree level, major & institution

Module E: Side-by-Side Candidate Comparison Studio

Enables recruiters to select between 2 and 4 applicants and view them in a synchronized matrix comparing match score breakdowns, skill coverage, total experience, educational credentials, and missing prerequisites to facilitate balanced hiring discussions.

Module F: 7-Stage Drag-and-Drop Recruitment Kanban Board

Organizes applicants across 7 pipeline columns: New -> Screening -> Shortlisted -> Interview -> Offer -> Hired -> Rejected. Features smooth HTML5 drag-and-drop, multi-candidate bulk actions, candidate counts by stage, stage duration metrics, and an instantaneous Undo button to reverse accidental transitions.

Module G: Talent Pool & Faceted Search

Enables recruitment teams to search historical applicants across all past postings using full-text keyword indexing, minimum experience sliders, categorized skill pills, and saved search query presets.

Module H: Gap-Targeted Interview Intelligence

Analyzes missing mandatory skills and weak competency areas for shortlisted candidates, dynamically generating tailored technical and behavioral interview questions. Interviewers can grade candidates using a standardized 1-5 evaluation rubric (Technical Competency, Problem Solving, Communication, and Team Fit).

Module I: Reports, Analytics & CSV Export

Generates recruitment funnel conversion charts, skill distribution graphs across candidate pools, and exports filtered candidate rankings to standardized CSV spreadsheets.

Module J: Privacy, GDPR & Blind Screening

Features an unconscious-bias toggle ("Blind Screening") that hides candidate names, photos, emails, phone numbers, and gender markers during initial screening. Includes automated data retention rules (90/180/365 days) and GDPR Right-to-be-Forgotten candidate record purging.

4. Technical Architecture & Zero-Dependency Design

Layer	Technologies	Purpose & Resilience
Frontend	React 18, Vite, Tailwind CSS, R...	20 client routes, responsive ed...
Backend API	Node.js v20, Express.js, Multer	12 REST controllers, cookie ses...
Database	MongoDB Atlas + MongoMemoryServer	Dual-mode: cloud cluster or loc...
AI Engine	Pluggable: Hugging Face / Local...	Sentence vector cosine similari...
Validation	Zod, bcryptjs, SHA-256 Crypto	Payload validation, password ha...

5. Verification Test Suite Results

The automated backend test suite (server/tests/runTests.js) asserts core algorithmic correctness across 5 test suites:

Test Suite	Function Under Test	Specification	Status
SHA-256 Fingerprinting	calculateHash()	Duplicate file detection & hash...	100% PASSED
Skill Extractor	extractSkills()	Boundary detection of C++, C#, ...	100% PASSED
Explainable Matching	calculateMatchScore()	4-component weighted formula & ...	100% PASSED
Job Readiness Analyzer	analyzeJobDescription()	Readiness score & buzzword iden...	100% PASSED
Vector Cosine Engine	calculateCosineSimilarity()	Semantic text vector distance c...	100% PASSED

6. Quick-Start Execution Guide

> **Pre-Seeded Recruiter Credentials**
> URL: http://localhost:5173 | Email: recruiter@hiresense.ai | Password: Password123!

Follow these commands to boot the complete application locally:

- **Start Backend API: cd c:\HCL_TRAINING\server && npm run dev (Runs on port 5000)**
- **Start Frontend Client: cd c:\HCL_TRAINING\client && npm run dev (Runs on port 5173)**
- **Run Automated Tests: cd c:\HCL_TRAINING\server && node tests/runTests.js**

7. Source Code & GitHub Repository

The entire source tree, frontend components, server controllers, seed data, sample resumes, and automated test suite have been initialized, committed, and pushed to GitHub:

- **Repository Link: https://github.com/VaibhavS18S/hiredsense.git**
- **Active Branch: main**
- **Sample Resumes for Testing: Located in sample_resumes/ (alex_morgan_fullstack.txt,** sarah_jenkins_devops.txt, marcus_reed_designer.txt)